

Cisco Enterprise Lab Modernization

Running a CCIE-Grade Networking Lab on IBM Cloud & Red Hat OpenShift

A Consulting Review of the CCIE Enterprise Infrastructure (v1.1) Lab Documentation
and the i3 Smart Labs Vendor Environments Matrix

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1. Executive Summary

This report reviews your existing Cisco CCIE Enterprise Infrastructure (EI) v1.1 lab documentation (Dell R620 / T5610 physical servers, 128–256 GB RAM, 1 TB SSD, running VMware ESXi 7.0 with EVE-NG 6.2.0, CML-2 2.9.1, SD-WAN 20.12.1 controllers, Cisco ISE 3.1, and CAT8000v virtual routers) and your cross-vendor i3 Smart Labs matrix, then defines how to evolve that environment into a modern, cloud-native lab platform hosted on **IBM Cloud** with **Red Hat OpenShift** as the orchestration layer.

Key conclusions:

- **The physical lab is functionally sound but architecturally legacy.** Single-server ESXi designs create a scaling ceiling, no GitOps/automation integration, and fragile image sprawl across EVE-NG/CML/individual CAT8000v VMs.
- **Keep the physical kit as a low-cost 'on-prem anchor', but move the build/CI/test pipeline to IBM Cloud + OpenShift.** OpenShift Virtualization (KubeVirt) can host EVE-NG/CML VMs as first-class citizens, while containerized tools (Containerlab, GNS3, Ansible, Batfish, NetBox) run natively on OpenShift.
- **Borrow proven patterns from other vendors** — Juniper's free cRPD/vJunos containers and Apstra intent-based networking, Palo Alto's VM-Series trials and Prisma SD-WAN design language, Fortinet's always-on free FortiGate-VM evals and overlay templates, F5 BIG-IP Next's container-first redesign, Arista's free cEOS-Lab and CloudVision/NetDL image distribution, and Aruba's AOS-CX free simulators.
- **A multi-emulator strategy is required:** Cisco CML (licensed, the only fully legal source of Cisco NX-OSv/CSR/IOS-XE images for labs), EVE-NG Community (free orchestration), Containerlab + vnetlab (open-source containerization of vendor VMs), GNS3 (legacy fallback), plus pure open-source NOS (FRRouting, VyOS, Open vSwitch) for scale and cost-free expansion.
- **Build a GitOps network pipeline on OpenShift:** Git (ArgoCD) → automated config generation (Ansible/Netmiko/Nornir) → pre-change validation (Batfish/Suzieq) → deploy to emulated fabric → telemetry verification (Prometheus/gNMI) → evidence capture. This turns a 'study lab' into an engineering asset.

Recommended target state: **Hybrid Lab Fabric** — physical Dell anchor nodes + IBM Cloud VPC compute (bare metal or nested virtualization-capable instances) + OpenShift Virtualization VM farm + containerized multi-vendor NOS estate, all fronted by a single lab-portal (NetBox as source of truth, Containerlab for topology definitions, GitOps for everything as code).

Detailed recommendations follow in Sections 2–11, with a costed bill of materials, an 18-month roadmap, and a risk register.

2. Review of the Existing CCIE Enterprise Infrastructure Lab

2.1 What the current design does well

- **Right-sized hardware:** Dell R620 with 2×10-core Xeon E5-2660v2 (40 vCPUs), 128–256 GB RAM, 1 TB SSD on PERC RAID gives genuine headroom for 10× CAT8000v + EVE-NG + CML + ISE concurrently.
- **Complete software stack for CCIE EI v1.1:** EVE-NG 6.2.0 with Viptela SD-WAN 20.12.1 controllers (vManage/vBond/vSmart), CML-2 2.9.1, ISE 3.1 (90-day demo), NSO on Ubuntu 20.04, Windows Server 2022, and 10 independent CAT8000v VMs.
- **Training assets bundled:** INE topology lab exercises, bootcamp videos, and initial configs — reduces time-to-lab for candidates.
- **Upgrade path documented:** RAM to 384 GB, SSD to 2/4 TB on R620 — a sensible mid-life extension.
- **Commercial protections:** replacement warranty, 4–5 day build time, continental-US ground shipping.

2.2 Gaps and risks identified

#	Gap / Risk	Impact	Severity
G1	Single physical server = single point of failure; no HA for lab services (vManage, ISE, CML)	Candidate downtime before exam windows; shared-lab contention	High

#	Gap / Risk	Impact	Severity
G2	ESXi 7.0 on end-of-support hardware generation (R620 = 2012 platform); no lifecycle plan	Firmware/CPU vulnerabilities; parts scarcity	Medium
G3	ISE is a 90-day demo and CML-2 is a 5-node eval — both require periodic re-licensing or repurchase	Interrupted study; license management burden	Medium
G4	No source of truth (IPAM/DCIM), no config versioning, no CI/CD — every topology is hand-built	Slow rebuilds, config drift, no repeatability	High
G5	Image management is manual across EVE-NG, CML, and standalone VMs	Version skew; hours lost per image update	Medium
G6	Licensing terms of pre-built vendor images (CAT8000v etc.) unclear from listing	Compliance exposure when used beyond personal study	High
G7	No multi-vendor exposure — only Cisco	Weakens designs that must interop (the exact reason your vendor matrix exists)	Medium
G8	No telemetry/observability stack (no Prometheus/Grafana, no flow analytics)	Cannot practice modern assurance skills	Low

2.3 Recommendations on the physical lab

- Keep the **R620 256 GB** variant as the on-prem anchor (best RAM ceiling). Repurpose the T5610 128 GB as a dedicated **EVE-NG Community** node if budget allows, or sell it.
- Move EVE-NG Community (free) onto it and retire the pre-built EVE-NG VM image — rebuild it from your own CCO-downloaded images for compliance and currency.
- Add a cheap 1–2 TB NVMe mirror for image storage and snapshot the ESXi datastores (ghettoVCB / Veeam Community) before every major build.
- Treat CML as the **licensed Cisco image factory** (buy the 1-year \$200 CML license through Cisco directly, never gray-market), and treat EVE-NG as the multi-vendor orchestration layer.
- Do **not** rely on the bundled ISE 90-day demo for anything but study; run production-grade AAA in the cloud with per-use tokens via Cisco dCloud instead.

3. Target Architecture — IBM Cloud + Red Hat OpenShift

Your stated platform is **IBM Cloud** with **Red Hat OpenShift**. This is a strong choice: IBM Cloud offers bare-metal and nested-virtualization-capable instances, VPC networking, Schematics (Terraform), and hybrid connectivity; OpenShift adds OpenShift Virtualization (KubeVirt) so VMs and containers share one platform, plus native GitOps (ArgoCD/Red Hat GitOps), Operators, and CI (OpenShift Pipelines/Tekton).

3.1 Three deployment patterns (pick per workload)

Pattern	What runs there	Pros	Cons
A. OpenShift Virtualization VM farm	EVE-NG Community VM, CML-2 VM, ISE, vManage/vSmart/vBond, CAT8000v/CSR/NX-OSv images	Native VM lifecycle on K8s; VM templates & snapshots; same GitOps for VMs and pods	Requires nested virtualization node (bare-metal OpenShift or instances with VT-x exposure)
B. Containerized NOS estate (Containerlab)	Arista cEOS-Lab, Juniper cRPD/vJunos-switch, Nokia SR Linux, FRRouting, VyOS, Open vSwitch, Cisco XRd/CSR containers via vnetlab	Seconds to spin a 50-node fabric; tiny footprint; topology-as-code (YAML)	Not all Cisco images are officially containerized; some vendors only ship VMs
C. Bare-metal / nested hosts on IBM Cloud	Large EVE-NG or CML server for the heaviest CCIE topologies (10+ CAT8000v + controllers)	Full bare-metal performance; exact exam-like conditions	Cost; cloud egress fees if you pull images nightly

3.2 Reference logical architecture

```
USERS (engineers / candidates / CI bots)
| HTTPS + console
+-----+
| IBM Cloud VPC (lab-vpc, 10.240.0.0/16) |
| ██████████ ██████████ ████████████████████
| | OpenShift | | OpenShift | | Bastion / | |
| | Cluster #1 | | Virtualization █ | VPN / jumphost █ |
| █ GitOps+CI █ █ node pool █ █ (wireguard) █ |
| █ ArgoCD █ █ ██████████ █ ██████████████████ |
| █ Tekton █ █ █ EVE-NG VM █ █ █ |
| █ NetBox(pod) █ █ █ CML-2 VM █ █ On-prem Dell R620 |
| █ Containerlab █ █ █ ISE VM █ █ anchor (EVE-NG) |
| ██████████████████ █ █ CAT8000v █ █ █ |
| █ ██████████ █ Site-to-site / |
| Object Storage ██████████████████████ Direct Link / |
| (image library, bare-metal nodes IPsec VPN |
| snapshots, ISOs) for CML/EVE-NG <████████████████████
+-----+
Observability: Prometheus + Grafana + Loki + SuzyBorg/Suzieq
Secrets: OpenShift Secrets + HashiCorp Vault (EaaS)
Automation: Ansible AWX/EDA (Event-Driven Ansible) on OpenShift
```

3.3 Sizing guidance (IBM Cloud)

Component	Suggested spec	Notes
OpenShift cluster (control+workers)	3× worker: 16 vCPU / 64 GB Gen2	Run GitOps, NetBox, AWX, Containerlab pods
Virtualization node pool	2× bare-metal or 32+ vCPU / 256 GB, nested virt enabled	Hosts EVE-NG/CML VMs and heavy CAT8000v workloads
Cloud Object Storage	1–2 TB	Vendor image library (cEOS, cRPD, vJunos, CSR, CAT8000v NX-OSv), topology snapshots, class recordings
VPC networking + VPN	2 subnets (mgmt/fabric), Site-to-site VPN or Direct Link Connect	Tunnel to on-prem R620; keep lab traffic off corporate WAN
Optional: Power Systems LPAR	IBM i/AIX partition	Only if your matrix's IBM track matters (AIX/IBM i skills)

Component	Suggested spec	Notes
Optional: GPU pool (NVIDIA track)	L4/A100 slices	Only for the AI/ML badge tracks in your vendor matrix

3.4 Why OpenShift specifically (vs plain RHEL/KVM)

- **OpenShift Virtualization:** treat EVE-NG/CML/ISE as VMs with K8s-native lifecycle, snapshots via VolumeSnapshots, and template-based cloning — instant 'class reset' after each cohort.
- **Red Hat GitOps (ArgoCD):** every topology YAML, Containerlab file, Ansible playbook, and NetBox seed data lives in Git; the cluster reconciles drift automatically.
- **Operators:** certified operators for Kafka (telemetry bus), Prometheus (monitoring), Keycloak (AAA for the lab portal), NetBox community operator.
- **Network plumbing:** Multus CNI + SR-IOV/bridge networks let you attach emulator pods to isolated L2 domains — essential for building multi-segment fabrics in-container.
- **Namespaces per team/exam-track:** RBAC quota isolation so CCIE, CCNP, security, and DevNet tracks never collide.

4. Ideas to Borrow from Other Vendors

The genius of your vendor matrix is that every vendor solved the same problem — 'how do we give people a lab?' — differently. Below are the transferable ideas, mapped to Cisco action.

4.1 Palo Alto Networks (Security track)

Their idea	What it is	Cisco lab adaptation
VM-Series trial VMs	Full firewall image, 30-day eval, no cost	Cisco FTDv/ASAv evals via CCO; publish both into one Containerlab fabric security + routing interop scenarios
Panorama central mgmt	Single pane for firewalls	Cisco Catalyst Center / vManage equivalent — use vManage 20.12.1 templates as your 'single pane' for SD-WAN studies
Prisma SD-WAN (CloudGenix) design language	App-aware, cloud-first WAN	Mirror it: run SD-WAN overlays where the vEdges sit behind commodity FRR/Open vSwitch 'underlay' you control fully
Strata Cloud Manager API-first ops	Everything is REST	Force the same discipline: all lab changes go through Cisco vManage/ISE/NSO REST APIs, stored in Git
Minemeld open-source threat-intel feeds	Community feeds → firewall objects	Run Minemeld as a pod on OpenShift; feed Umbrella/SecureX/ISE profiling with synthesized threat data

4.2 Juniper Networks (Networking track)

Their idea	What it is	Cisco lab adaptation
Free cRPD container	Routing daemon in Docker, free	Create 'virtual route-reflector / route-server' pods to scale BGP topologies infinitely — Cisco devices peer with open-source reflectors
vJunos-switch / vJunosEvolved evals	Free virtual switching with account evals	Add a Junos leaf to your Cisco spine-leaf to practice true multi-vendor EVPN/VXLAN — the #1 real-world gap in CCIE study
Apstra intent-based networking	Graph-based design + validation	Replicate the *pattern* with Batfish (intent validation) + Suzieq (network digi twin) — open source, no license
OpenConfig/YANG everywhere	Model-driven management	Standardize lab automation on OpenConfig/gNMI where Cisco supports it (IOS-XR, NX-OS, CAT8K) instead of only CLI screenscraping
Junos 'commit confirmed'	Auto-rollback configs	Implement the same safety net in your Ansible playbooks (napalm_install_config with rollback) for every lab change

4.3 Fortinet (Security + SD-WAN track)

Their idea	What it is	Cisco lab adaptation
Always-free FortiGate-VM eval (limited throughput)	Permanent lab firewall	Stock one FTDv + one ASAv permanently; use them as the fixed security boundary of every topology
FortiManager templates & SD-WAN overlays	Zero-touch overlay templates	Copy the template-first approach: build vManage feature templates for CAT8000v once, reuse forever
Free FortiAnalyzer eval	Central logging	Stand up a Grafana+Loki or Splunk-free (Suzieq+GNMIIc) logging fabric; make every lab exercise end with a report pulled from telemetry
FORTIGATE REST API + Ansible collections	Day-0 automation	Mirror with Cisco collection (cisco.ios, cisco.nxos, cisco.sdwan) — every lab starts from an API call, not console

4.4 F5 Networks (Load balancing / ADC)

Their idea	What it is	Cisco lab adaptation
BIG-IP Next container-first redesign	ADC as containers/K8s operator	Cisco has no container ADC; practice the *pattern* with open-source HAProxy/Envoy on OpenShift and study F5-CA concepts separately
Free BIG-IP VE trial	30-day full ADC	Add to interop topologies as the load-balancer in front of redundant CSR/CAT8K services
Declarative AS3 (JSON) configs	App services as code	Adopt the same JSON-as-source pattern for Cisco: store vManage policies, ISE authorization profiles, and NSO services as versioned JSON/YAML

4.5 Arista Networks (Cloud networking track)

Their idea	What it is	Cisco lab adaptation
cEOS-Lab free container	Official Arista EOS as a Docker image, free with account	The best free multi-vendor partner for Cisco: EOS + NX-OSv + IOS-XE in on EVPN fabric via Containerlab
NetDL (software portal)	Single portal for all images & licenses	Build your internal 'NetDL': an OpenShift-hosted artifact registry (Harbor) + NetBox catalog so engineers self-serve images
CloudVision & Telemetry-first ops	Streaming gNMI from day one	Make gNMI telemetry mandatory in the lab: every device streams to Prometheus; no exercise is complete without a dashboard
Config replace & 'show diff' culture	Safe declarative changes	Implement config-replace workflows on IOS-XE/IOS-XR labs so students learn rollback discipline

4.6 Aruba (HPE) & Check Point & others — quick hits

Vendor	Borrow this idea	Cisco lab adaptation
Aruba AOS-CX	Free AOS-CX OVA simulator + CX Studio	Practice OSPF/BGP/EVPN interop between AOS-CX and NX-OSv/CAT8K
Check Point CloudGuard	Policy-as-code (Terraform provider)	Manage Cisco FTD/ASA policies via Terraform (fmc provider) stored in Git
VMware	Nested ESXi / VCSA labs	You already have ESXi — the lesson is nested labs; replicate: OpenShift Virt nodes inside OpenShift Virt for disposable 'ESXi-in-ESXi' style sandboxes
Nutanix	CE (Community Edition) free cluster	Pattern: a free hypervisor tier for the lab — your OpenShift Virtualization nodes are the equivalent
AWS (LocalStack)	Local emulation of cloud APIs	Cisco analog: Cisco Modeling Labs 'external connector' nodes + dCloud sandboxes emulate WAN/cloud attachments locally
Red Hat	OpenShift Sandbox / Developer Sandbox	Offer a 'Network Engineer Sandbox' namespace preloaded with Containerlab, NetBox, AWX — self-service lab vending
IBM	Power LPARs, Schematics IaC	Use IBM Schematics (Terraform) to build/destroy the entire cloud lab platform on a schedule — huge cost saver (nightly destroy)
Splunk/SolarWinds	Free trials as the observability baseline	Prometheus/Grafana/Loki are the free permanent replacement; trials only for exposure

5. Open-Source Tools & Emulator Catalog

5.1 Emulator / NOS platforms (multi-vendor)

Tool	Type	License	Runs on OpenShift?	Best for
Containerlab	Container topology orchestrator (YAML)	Free, open source (Apache 2.0)	Yes — native pods, use privileged + Multus	First choice for all multi-vendor fabrics; seconds to boot; topology-as-code
vrnetlab / vr-xcon	Convert vendor VMs (qcow2) to containers	Free, open source	Yes (privileged pods)	Cisco CSR1000v, XRv9k, NX-OSv, Juniper vMX, FortiGate-VM as containers
GNS3	Desktop/server network emulator	Free, open source	Server component yes	Legacy compatibility, GUI-driven exploration
EVE-NG Community	VM-based multi-vendor emulator	Free community edition	As a VM (OpenShift Virt)	Your current skills base; multi-vendor images in one UI
Cisco Modeling Labs (CML)	Cisco's official emulator	Paid (\$200/yr personal)	As a VM	Only fully legitimate source of NX-OSv 9000, CSR1000v, IOS-XRv9k, CAT8K images
FRRouting (FRR)	Open-source routing suite (BGP/OSPF/IS-IS/PIM/EVPN)	Free, open source	Yes — tiny container	Massive scale nodes, route-reflectors, underlay control
VyOS	Open-source router/firewall NOS	Free rolling release	Yes (container/VM)	Stand-in CE/customer edge devices
Open vSwitch (OVS)	Open-source virtual switch	Free, open source	Yes	L2 stitching between pods/VMs; dataplane plumbing
OpenBSD / Linux netem	Traffic shaping & impairment	Free	Yes	WAN impairment labs (latency/loss) for SD-WAN policy testing
Trex / Ostinato	Traffic generation	Free, open source	Yes	Throughput & QoS verification exercises
Mininet	SDN prototyping	Free, open source	Yes	APIC-EM/SDN-era concepts, OpenFlow basics
NVIDIA Containerlab kind + SR Linux	Free container NOS	Nokia SR Linux free container	Yes	State-of-the-art EVPN practice alongside Cisco

5.2 Automation, validation & observability (open source)

Tool	Purpose	Why it matters to the Cisco lab
NetBox	IPAM/DCIM source of truth	Single inventory for all topologies; feeds Ansible/Containerlab
Ansible + AWX/EDA	Config management & event-driven ops	Official Cisco collections (cisco.ios, cisco.nxos, cisco.sdwan, cisco.ise)
Nornir / Netmiko / Scrapli	Python-native network automation	Fast, threaded device interaction for custom tooling
Batfish	Vendor-independent config analysis	Pre-change validation: 'will this ACL/BGP change break reachability?'
Suzieq	Network observability/digital twin	Multi-vendor 'show commands at scale' with REST API
Batfish + cisco data model	Cisco-aware analysis	Validates IOS-XE/IOS-XR/NX-OS configs before deployment
gNMIc / gRPC telemetry	Streaming telemetry client	Collects from CAT8000v, NX-OSv, IOS-XR into Prometheus
Prometheus + Grafana + Loki	Metrics, dashboards, logs	Your free 'Catalyst Center + DNAC assurance' replacement
Tinker / TeraFlow? (skip)	—	—
Keycloak	Identity for the lab portal	SSO for NetBox, AWX, Grafana with LDAP/AD federation
Harbor	Private container/image registry	Internal NetDL: stores cEOS, cRPD, SR Linux, FRR images + qcow2 conversion pipeline
ArgoCD + Tekton	GitOps & CI	Topology promotion pipeline: dev → staging → 'exam-like' prod

5.3 Cisco images — legitimate acquisition paths

Cisco image/VM	Legitimate source	Cost
CAT8000v (Catalyst 8000V)	Cisco CML 2.x node definitions / Cisco dCloud	Included with CML license

Cisco image/VM	Legitimate source	Cost
CSR1000v	CML, or AWS/Azure marketplace (BYOL/trial)	CML included; cloud hourly ~\$0.05–0.30
NX-OSv 9000	CML only	CML included
IOS XRv 9000	CML only	CML included
IOSv / IOSvL2	CML	CML included
ASAv / FTDv	CCO eval (30-day) via Cisco.com	Free eval
vManage/vSmart/vBond 20.12.x	Cisco SD-WAN eval (dCloud) or Viptela evaluation	Free trial
Cisco ISE	90-day demo (CCO)	Free demo
APIC-EM / Catalyst Center	dCloud reservation	Free
Cisco dCloud sandboxes	dcloud.cisco.com	Free with CCO

Compliance note: only images obtained via CCO/CML/dCloud or official vendor eval portals should be deployed. Gray-market pre-built labs (the source of the reviewed document) are convenient but create licensing exposure if the environment is used commercially or by multiple employees. Your target design eliminates that exposure because every image comes from a verifiable source.

6. GitOps Reference Pipeline (Network-as-Code on OpenShift)

```

engineer push ■■■ GitLab/GitHub repo
■ ■■ topologies/ (Containerlab .clab.yml)
■ ■■ netbox/ (seed data)
■ ■■ ansible/ (playbooks, inventories)
■ ■■ policies/ (vManage, ISE, FMC JSON)
▼
Tekton CI: lint → secrets-inject (Vault) → Batfish validate
▼ (green build)
ArgoCD sync to lab namespace (dev → staging → prod)
▼
Containerlab deploys fabric from Harbor images (private registry)
▼
AWX/EDA runs day-0/day-1 (Ansible Cisco collections)
▼
gNMic → Prometheus → Grafana dashboards; Suzieq snapshot
▼
Verification gate (pytest-network + Batfish reachability)
▼
Evidence pack (logs + configs + dashboard PDF) pushed to Object Storage
▼
ArgoCD auto-destroy at 02:00 (IBM Schematics TF destroy = cost control)

```

Key design decisions:

- **Topology-as-code first:** if a fabric can't be expressed as a Containerlab YAML, it doesn't exist. Manual point-and-click topologies are banned from the shared platform.
- **Namespaces per track:** ccie-ei, ccnp, security, devnet — each with its own ArgoCD Application and ResourceQuota.
- **Immutable images:** Harbor stores pinned image digests; 'latest' tags are blocked in prod namespaces.
- **Event-Driven Ansible:** telemetry alarm (interface down, BGP flap in a lab) triggers auto-remediation playbooks — your students learn modern ops, not just CLI.
- **Cost guardrails:** all cloud resources tagged; a nightly IBM Schematics/Terraform destroy job for non-prod namespaces; budget alerts at 70/90%.

7. Bill of Materials & Cost Estimate (USD, indicative)

Item	Spec / License	One-time	Monthly	Source
On-prem anchor (existing)	Dell R620 256GB — reuse, add 2TB NVMe	\$0 (sunk)	\$0	Your current lab
CML license	1-year Cisco Modeling Labs personal	\$200/yr	\$17	Cisco
IBM Cloud VPC workers	3× 16vCPU/64GB Gen2	—	~\$620	IBM Cloud catalog
Virtualization node pool	2× 32vCPU/256GB (nested virt)	—	~\$1,100	IBM Cloud
Object Storage	2 TB + egress	—	~\$45	IBM Cloud COS
VPN/Direct Link	IPSec VPN (or DL Connect)	—	\$0–500	IBM Cloud
OpenShift	Included via IBM RHOCPS service or self-managed on RHEL	—	license-dependent	IBM/Red Hat
Open-source stack	Containerlab, NetBox, AWX, Harbor, Prometheus, Grafana, Batfish, Suzieq, Keycloak, ArgoCD, Tekton	\$0	\$0 (ops time)	Community
Vendor eval images	cEOS-Lab (free w/ account), cRPD (free), SR Linux (free), vJunos (free), FortiGate-VM eval, FTDv eval	\$0	\$0	Vendor portals

Item	Spec / License	One-time	Monthly	Source
Optional: bare-metal CML host	1× IBM Cloud bare metal 32C/256GB	—	~\$900 (only when needed)	IBM Cloud bare metal

Steady-state monthly estimate: **\$1,800–2,400** for the full cloud platform (before OpenShift subscription). With scheduled destroy of non-prod namespaces outside business hours, effective compute cost drops ~40%. The on-prem R620 handles burst CCIE-scale scenarios at zero marginal cost.

8. 18-Month Implementation Roadmap

Phase	Timeline	Key deliverables	Exit criteria
0. Stabilize	Month 0–1	Rebuild EVE-NG on R620 from CCO/CML images; snapshot ESXi; buy CML license; inventory images in NetBox	All current CCIE exercises run on compliant images; restore test < 30 min
1. Cloud landing zone	Month 1–3	IBM Cloud VPC + VPN; OpenShift cluster; Harbor, Keycloak, NetBox, Prometheus/Grafana/Loki stack	Platform accessible; SSO works; observability sees lab devices
2. Virtualization farm	Month 3–5	OpenShift Virt node pool; EVE-NG & CML as VM templates; volume snapshot automation; CAT8000v golden images	1-click deploy of full CCIE EI topology from template in < 20 min
3. Container fabrics	Month 4–7	Containerlab operator + Multus; cEOS/cRPD/SR-Linux/FRR/VyOS registry; first multi-vendor EVPN lab (Cisco spine + Junos/EOS leafs)	50-node fabric boots in < 5 min; telemetry dashboards live
4. GitOps pipeline	Month 6–9	ArgoCD + Tekton + AWX/EDA; Batfish/Suzieq gates; vManage/ISE/FMC policy-as-code repos	Every change via PR with pre-check evidence; zero manual prod changes
5. Vendor interop tracks	Month 8–12	Fortinet/CheckPoint security pods; F5 VE trial track; Aruba AOS-CX sim; AWS/GCP local attachment nodes; IBM Power LPAR (optional)	4 new interop courses published using same pipeline
6. Productize	Month 12–18	Self-service lab portal (Backstage) with quota vending; scheduled environments; cost dashboard; DR plan for on-prem anchor	Engineers self-serve labs; monthly cloud cost per active user reported

9. Risk Register & Mitigations

Risk	Likelihood	Impact	Mitigation
Nested virtualization performance penalty on cloud instances	Medium	Medium	Use bare-metal OpenShift nodes for the CML/EVE farm; benchmark before migrating
OpenShift Virt + emulator privileged pods = security exposure	Medium	High	Dedicated node pool, SELinux enforcing, seccomp profiles, network policy default-deny, no prod corp connectivity
Vendor eval licenses expire mid-course	High	Low	Automation re-arms demos; maintain image/license calendar in NetBox; fallback to open-source NOS
Cisco image licensing scrutiny	Medium	High	100% CCO/CML/dCloud-sourced images; document entitlements; avoid gray-market pre-built labs in shared infra
Cloud egress costs (telemetry/images)	Medium	Medium	Keep telemetry local to VPC; compress snapshots; lifecycle policies on COS
Skills gap: team knows ESXi, not OpenShift/KubeVirt	High	Medium	Red Hat training (DO280/DO316), pair-programming phase, runbooks, start with pattern A only
Lab-portal sprawl (too many tools)	Medium	Medium	Backstage single pane; consolidate on Containerlab+NetBox+AWX trinity; deprecate GNS3 after migration

10. Top 10 Quick Wins (do these first)

- 1. Buy the official CML license and rebuild one clean image library — single biggest compliance + currency win.
- 2. Stand up NetBox (container on OpenShift or the R620) and inventory every node, IP, image, and license today.
- 3. Deploy Containerlab on a single Linux VM with cEOS-Lab + FRRouting — first multi-vendor fabric in a week, \$0.
- 4. Put all existing INE initial configs into Git; make Ansible load them via napalm_install_config with auto-rollback.
- 5. Add gNMIc → Prometheus → Grafana for the CAT8000v fleet; one dashboard beats zero assurance skills.

- 6. Wire Batfish into the Ansible flow so every lab change gets a reachability check before commit.
- 7. Reserve Cisco dCloud SD-WAN/ISE sandboxes for the heavy controller labs instead of self-hosting eval VMs.
- 8. Enable EVE-NG VM snapshots on ESXi; scripted restore between candidate sessions.
- 9. Tag every IBM Cloud resource and set a budget alert — cost discipline from day one.
- 10. Write the lab's 'definition of done': topology YAML + configs + telemetry dashboard + evidence pack in Git.

11. Appendix — Minimal Working Snippets

Containerlab: Cisco CAT8000v (via vrnetlab) + Arista cEOS + FRR EVPN fabric

```
name: ccie-fabric
topology:
nodes:
spine1: {kind: cisco_cat8000v, image: vrnetlab/cisco_cat8000v:17.12.01}
leaf-eos: {kind: arista_ceos, image: ceos:4.33.0F}
leaf-frr: {kind: linux, image: frrouting/frr:latest}
host1: {kind: linux, image: alpine:latest}
links: [spine1:eth1-leaf-eos:eth1, spine1:eth2-leaf-frr:eth1, leaf-eos:eth2-host1:eth1]
```

gNMIc telemetry scrape for CAT8000v

```
gnmic -a 10.240.1.11:57400 -u admin -p $VAULT_PASS \
--insecure subscribe \
--path /interfaces/interface/state/oper-status \
--path /bgp/neighbors/neighbor/state/session-state -o prometheus
```

Ansible: compliant config replace with rollback (Junos-style safety)

```
- hosts: cat8k
tasks:
- cisco.ios.ios_config:
src: '{{ topology }}_{{ inventory_hostname }}.cfg'
replace: config # atomic replace, not merge
register: r
- assert: { that: 'not r.changed or batfish_pre_check_passed' }
```

IBM Cloud Schematics: nightly cost control

```
# cron in OpenShift (CronJob) calls Schematics workspace actions:
# weekdays 07:00 → terraform apply (lab namespaces)
# weekdays 02:00 → terraform destroy (non-prod namespaces)
```

— End of document. Contact the network consulting team for detailed design sessions per phase.